

BONDING – IONIC BONDING

KEY VOCABULARY

Ionic bond: the electrostatic attraction between oppositely charged ions.

Ion: a charged particle made when atoms lose or gain electrons.

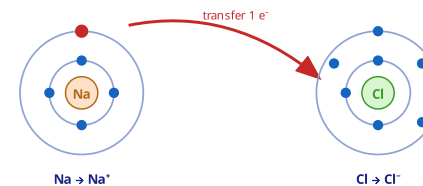
Cation: a positive ion (metal that has lost electrons).

Anion: a negative ion (non-metal that has gained electrons).

COMMON MISCONCEPTIONS

- ~~Electrons are shared in ionic bonding.~~
- ✓ Electrons are transferred, not shared.
- ~~Ionic compounds conduct when solid.~~
- ✓ They only conduct when molten or dissolved.
- ~~Metals gain electrons to form ions.~~
- ✓ Metals lose electrons to form positive ions.

ELECTRON TRANSFER (NA + CL)



1 DEFINE IONIC BONDING

State what is meant by an **ionic bond**. [2 marks]

2 FORMING IONS

Sodium loses one electron. State the **charge** of the ion formed and why. [2 marks]

3 CONDUCTIVITY

Explain why sodium chloride conducts electricity when molten but not when solid. [2 marks]

4 MELTING POINT

Explain why ionic compounds have high melting points. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

In ionic bonding electrons are ____ from the metal, which becomes a ____ ion.

transferred

shared

positive

negative

6 WORK OUT A FORMULA

Magnesium forms Mg²⁺ and oxygen forms O²⁻. Give the formula of magnesium oxide. [1 mark]
